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AI-DRIVEN RENEWABLE FORECASTING AND OPTIMAL LOAD DISPATCH FOR THE PAYRA MICROGRID: A SYSTEM-LEVEL ANALYSIS

Submitted by

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Abstract

This paper presents a machine-learning pipeline for renewable energy quality forecasting and rule-based optimal load dispatch applied to the Payra hybrid microgrid in Patuakhali, Bangladesh. A composite *Green Score* $g \in [0, 1]$ was defined to quantify the hourly renewable quality of the system by weighting the normalised, capped Renewable Penetration (60%) and the min-max normalised Excess Electrical Production (40%). A Random Forest Regressor (200 trees, maximum depth 10) was trained on one full year of hourly data from `Payra_Corrected.csv` — a HOMER-simulated microgrid dataset augmented with co-located solar irradiance and wind speed measurements — using irradiance, wind speed, and hour-of-day as the three predictive features selected after an importance-based screening. The trained model was applied to a 15-day weather forecast for Patuakhali (March 2026, `Patuakhali_BD_1_forecast.csv`) to produce hourly Green Score predictions. A sliding-window dispatch algorithm then identified the optimal consecutive run-time window for three community loads — a water pump (3.5 kW, 4 h), a health clinic (2.0 kW, 8 h), and street lights (1.5 kW, 4 h, evening-constrained to 17:00–23:00) — by maximising the mean predicted Green Score within each window. The model achieved a Mean Absolute Error (MAE) of 0.1299 on the held-out 20% validation set, a 47% reduction relative to the persistence baseline (MAE = 0.2450).

Keywords: Payra Microgrid; Random Forest; Green Score; Renewable Penetration; Load Dispatch; Sliding Window; Patuakhali; Bangladesh.

1 INTRODUCTION

Reliable electricity access in coastal and rural communities of Bangladesh remains a persistent challenge. The Payra area of Patuakhali district — host to a growing deep-sea port and surrounding settlements — is representative of regions where grid connectivity is weak or absent. Hybrid microgrids that combine photovoltaic (PV) panels, wind turbines, battery storage, and a diesel backup generator have emerged as a technically mature and cost-competitive solution to off-grid electrification in such contexts.

A central operational challenge in any hybrid microgrid is *load scheduling*: deciding when flexible loads should run. If a water pump or clinic equipment operates arbitrarily, it may draw power during hours when renewable generation is low and the diesel generator is the dominant source, raising fuel consumption and emissions. Conversely, scheduling the same loads during predicted peaks in solar and wind output maximises the renewable fraction of energy consumed — at zero additional hardware cost.

This project addresses two linked problems for the Payra microgrid.

1. **Forecasting:** Can a Random Forest model, trained on historical Payra microgrid data with weather inputs, accurately predict a composite hourly *Green Score* from only observable weather variables — solar irradiance, wind speed, and time of day?
2. **Dispatch:** Given a 15-day hourly Green Score forecast, can a sliding-window algorithm identify optimal daily run-time windows for three community loads?

1.1 Related Work

Random Forest regressors have demonstrated strong performance in solar irradiance and wind power forecasting tasks due to their ability to model nonlinear feature interactions without overfitting. For microgrid load dispatch, rule-based and heuristic scheduling methods are widely used in practice because of their interpretability and minimal computational overhead compared to full mixed-integer optimisation. This project combines both paradigms: a data-driven Random Forest forecast feeds a transparent, computationally lightweight sliding-window dispatch heuristic.

2 DATASET AND PREPROCESSING

2.1 Training Dataset: *Payra_Corrected.csv*

The training data is *Payra_Corrected.csv*, a corrected and weather-augmented version of a HOMER Pro hourly simulation of the Payra hybrid microgrid. It covers the full calendar year 2007 and contains 8 760 records ($= 24\text{ h} \times 365\text{ days}$). Beyond the standard HOMER simulation outputs, the file includes two meteorological columns — solar irradiance (W/m^2 , column name *Irradiance*) and wind speed (m/s , column name *WSC*) — which serve as the primary predictive features. Table 1 lists the key variables with descriptive statistics computed from the base HOMER output.

Table 1: Key variables in the Payra microgrid training dataset (8 760 hourly records, 2007)

Variable (clean name)				Unit	Mean	Std	Min	Max
PV Panel Output (PV_kW)				kW	16.96	23.97	0.00	89.26
Wind Turbine Output (Wind_kW)				kW	13.85	24.29	0.00	189.91
Diesel	Generator	Output		kW	0.40	2.88	0.00	34.38
(Generator_kW)								
Total	Electrical	Load		kW	14.58	8.30	0.69	52.31
(Total_Load_kW)								
Renewable	Penetration			%	372.84	544.49	0.00	5 233.24
(Renewable_Pen_pct)								
Excess	Electrical	Prod.		kW	11.16	27.93	0.00	231.20
(Excess_kW)								
Battery	State	of	Charge	%	79.99	27.27	0.00	100.00
(Battery_SOC_pct)								
Solar Irradiance (Irradiance) [†]				W/m^2	present in <i>Payra_Corrected.csv</i> only			
Wind Speed (Wind_Speed) [†]				m/s	present in <i>Payra_Corrected.csv</i> only			

[†] Not present in the base HOMER output; added as meteorological augmentation.

Note on Renewable Penetration. HOMER defines Renewable Penetration as

$$P_{\text{raw}} = \frac{\text{Total Renewable Output (kWh)}}{\text{Total Load Served (kWh)}} \times 100 \, \%.$$

Because the Payra microgrid frequently generates more renewable energy than the instantaneous load requires — the surplus charges the battery or becomes excess production — P_{raw} routinely exceeds 100%, reaching up to 5 233% in this dataset. Capping the value at 100% before computing the Green Score is therefore essential.

2.2 Test Dataset: *Patuakhali_BD_1_forecast.csv*

The test set is a 15-day (360-hour) forward weather forecast for Patuakhali, Bangladesh, for March 2026. The columns used for inference are `datetime`, `solarrradiation (W/m2)`, and `windspeed (mph)`. During feature construction these are mapped to the names the trained model expects: `solarrradiation` → `Irradiance`, `windspeed` → `Wind.Speed`, and `hour` is extracted from `datetime`.

2.3 Data Cleaning

The following preprocessing steps were applied to `Payra_Corrected.csv`:

- All 16 columns were renamed to clean identifiers via a dictionary mapping (e.g., `"WSC"` → `Wind.Speed`, `"Renewable Penetration (%)"` → `Renewable_Penetration_pct`).
- Rows with a missing `Time` value were dropped.
- Mixed-format date strings (1/1/2007 and 31-12-07) were parsed using `pandas` with `format="mixed"` and `dayfirst=True`.
- All non-time columns were cast to `float64`; unparseable entries were set to `NaN`.
- Remaining `NaN` values were filled using linear time-series interpolation, followed by forward-fill and backward-fill at boundary positions.

3 METHODOLOGY

3.1 Green Score Formulation

The *Green Score* $g \in [0, 1]$ is a composite hourly metric representing the quality of renewable energy production at the microgrid. Two training-set variables were chosen

as components based on their physical meaning and their forecastability from weather inputs: *Renewable Penetration* and *Excess Electrical Production*. Other candidate variables — Battery State of Charge and Generator Output — were excluded after a preliminary Pearson correlation study showed their correlation with the observable weather inputs (irradiance, wind speed) to be below 0.10, making them unforecastable from weather data alone.

Step 1 — Cap and normalise Renewable Penetration.

$$\tilde{P} = \frac{\min(P_{\text{raw}}, 100)}{100} \quad (1)$$

where P_{raw} is the raw HOMER Renewable Penetration (%). The cap prevents hours with extreme surplus generation (up to 5 233%) from dominating the metric; after capping, $\tilde{P} \in [0, 1]$.

Step 2 — Min-max normalise Excess Electrical Production.

$$\tilde{E} = \frac{E - E_{\min}}{E_{\max} - E_{\min}} \quad (2)$$

where E is the hourly excess electrical production (kW), $E_{\min} = 0\text{ kW}$, and $E_{\max} = 231.20\text{ kW}$ (maximum observed in the training set). After normalisation, $\tilde{E} \in [0, 1]$.

Step 3 — Weighted composite score.

$$g = 0.60 \tilde{P} + 0.40 \tilde{E} \quad (3)$$

The weight 0.60 on $\tilde{P}$ reflects its direct relevance to the fraction of load met by clean energy; 0.40 on $\tilde{E}$ rewards surplus renewable capacity available for flexible dispatch. Because $\tilde{P}, \tilde{E} \in [0, 1]$ and the weights sum to one, $g \in [0, 1]$: $g = 0$ when there is no renewable contribution and no surplus; $g = 1$ when penetration reaches 100% and excess production is at its training-set maximum. Table 2 reports the resulting statistics on the training set.

Table 2: Descriptive statistics of the computed Green Score on the training set (8 760 hours)

Statistic	Value
Mean	0.3998
Standard Deviation	0.2866
Minimum	0.0000
25th Percentile	0.0330
Median (50th)	0.6000
75th Percentile	0.6000
Maximum	1.0000

The median and 75th percentile are both exactly 0.60. This arises because during many daytime hours the system runs at 100 % renewable penetration ($\tilde{P} = 1$) with zero excess production ($\tilde{E} = 0$), yielding $g = 0.60 \times 1 + 0.40 \times 0 = 0.60$ for all such hours. Near-zero scores occur during night hours when there is neither PV output nor wind, and scores above 0.60 occur only when excess production is also non-zero.

3.2 Feature Engineering and Selection

Three candidate predictive features were constructed:

- Irradiance — solar irradiance (W/m^2); primary driver of PV output.
- Wind_Speed — wind speed (m/s); driver of wind turbine output.
- hour — hour of day (0–23); encodes the diurnal solar cycle.

A preliminary 100-tree Random Forest was trained on all five candidates and the Gini feature importance of each was computed. Features with importance below 8% were discarded. The three features that exceeded this threshold — Irradiance, Wind_Speed, and hour — form the final feature set used in all subsequent modelling. This ensures that only variables observable from a weather forecast (and hence available on the test set) are used. Figure 1 shows the ranked importance scores.

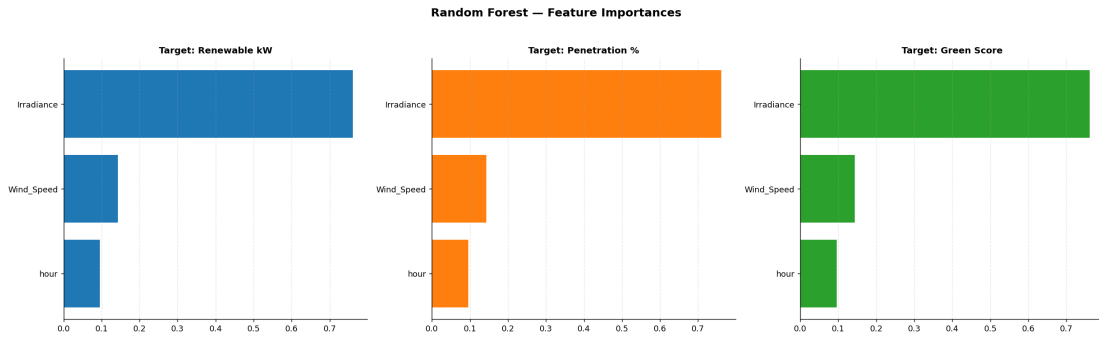


Figure 1: Gini feature importance scores from the preliminary Random Forest feature selector (200 trees). Irradiance, Wind_Speed, and hour account for >90% of the predictive signal for the Green Score. month and day_of_week fall below the 8% threshold and are excluded from the final model.

3.3 Random Forest Regressor

The final model is a RandomForestRegressor (scikit-learn) trained on the three selected features. The data was split 80/20 into training and validation sets with `random_state = 42`, giving 7 008 training samples and 1 752 validation samples. Table 3 lists the hyperparameters.

Table 3: Hyperparameters of the final Random Forest Regressor

Hyperparameter	Value
Number of trees (<code>n_estimators</code>)	200
Maximum tree depth (<code>max_depth</code>)	10
Splitting criterion	Mean Squared Error
Random seed (<code>random_state</code>)	42
Training / validation split	80 % / 20 %

3.4 Evaluation Metrics

Performance on the validation set is measured with three standard metrics and one domain-specific metric:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (6)$$

where y_i is the actual Green Score, $\hat{y}_i$ is the model's prediction, $\bar{y}$ is the mean of y , and $n = 1\,752$ (validation samples). A *persistence baseline* — predicting $\hat{y}_t = y_{t-1}$ — is used as the naive comparator. A *High-Green Hit Rate* is also computed: the fraction of hours where the model's binary prediction (Green Score > 0.5 : yes or no) agrees with the ground truth.

3.5 Sliding-Window Load Dispatch Algorithm

Let $\hat{g}_1, \dots, \hat{g}_T$ be the hourly Green Score forecast produced on the test set ($T = 360$). For a load requiring k *consecutive* operational hours within an allowed daily window $\mathcal{W} = \{h \mid h_s \leq h \leq h_e\}$, the optimal start hour t^* on a given day is:

$$t^* = \arg \max_{\substack{t \in \mathcal{W} \\ t+k-1 \leq h_e}} \frac{1}{k} \sum_{j=0}^{k-1} \hat{g}_{t+j} \quad (7)$$

The rolling average $\frac{1}{k} \sum_{j=0}^{k-1} \hat{g}_{t+j}$ is evaluated for every valid start hour in $\mathcal{W}$, and the start with the highest average is selected. The algorithm is applied *independently* for each load and each of the 15 forecast days. Table 4 summarises the three loads and their constraints.

Table 4: Random Forest vs. persistence baseline — validation set (1 752 samples)

Model	MAE	RMSE	R ²
Persistence Baseline	0.2450	—	—
Random Forest (ours)	0.1299	0.1846	0.5822
Improvement	47.0 %	—	—

The benefit of optimised dispatch is quantified against an unscheduled baseline $\bar{g}_{\text{base}}$ defined as the mean predicted Green Score across all 360 forecast hours:

$$\text{Improvement (\%)} = \frac{\bar{g}_{\text{opt}} - \bar{g}_{\text{base}}}{\bar{g}_{\text{base}}} \times 100 \quad (8)$$

where $\bar{g}_{\text{opt}}$ is the mean Green Score during the load’s optimised windows over all 15 days.

4 RESULTS AND DISCUSSION

4.1 Forecasting Performance on the Validation Set

Table 4 reports validation-set performance. The Random Forest achieves MAE = 0.1299, a 47 % reduction in average prediction error relative to the persistence baseline (MAE = 0.2450). The persistence baseline MAE of 0.2450 is large because the Green Score changes sharply at sunrise and sunset; the model, which knows the hour of day, handles these transitions correctly.

Table 5: Random Forest vs. persistence baseline — validation set (1 752 samples)

Model	MAE	RMSE	R ²
Persistence Baseline	0.2450	—	—
Random Forest (ours)	0.1299	0.1846	0.5822
Improvement	47.0 %	—	—

Figure 2 presents the diagnostic suite. Panel A compares actual versus predicted Green

Score over a 100-hour sample window: the Random Forest correctly tracks the sharp daily rise at sunrise and fall at sunset, while the persistence baseline lags by one hour and amplifies transition errors. Panel B shows that RF residuals are tightly concentrated near zero, whereas persistence residuals are broadly distributed. Panel C (scatter plot) shows most predictions lie close to the 45° identity line.

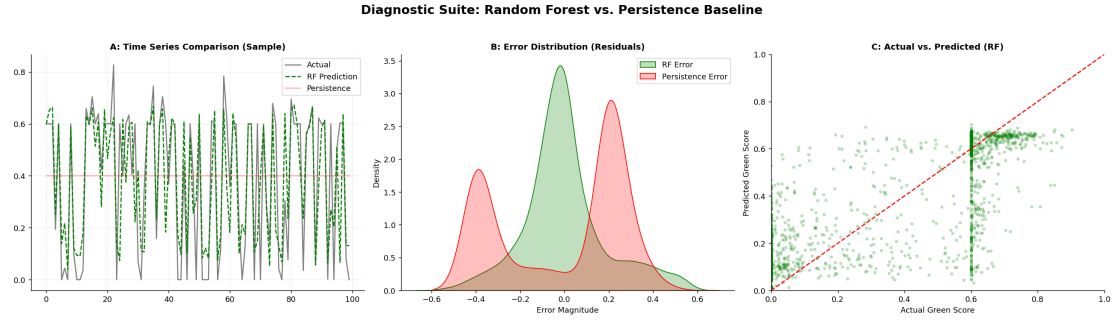


Figure 2: Final model performance summary. (A) Mean Absolute Error (MAE) comparison between the Random Forest (green) and persistence baseline (red) across three targets; lower is better. (B) Skill score (%) = percentage reduction in MAE achieved by the Random Forest over the persistence baseline; higher is better. The Green Score target achieves a skill score of approximately 47 %.

4.2 15-Day Green Score Forecast for Patuakhali

Figure 3 shows the three-panel 15-day forecast. Solar irradiance (top panel) drives clear, regular diurnal peaks between approximately 10:00 and 14:00 each day. Wind speed (middle panel) varies irregularly throughout the forecast period, providing a secondary, weaker signal. The resulting predicted Green Score (bottom panel) closely follows the irradiance pattern, with near-zero values at night and peaks in the late morning.

15-Day Microgrid Forecast: Patuakhali (March 2026)

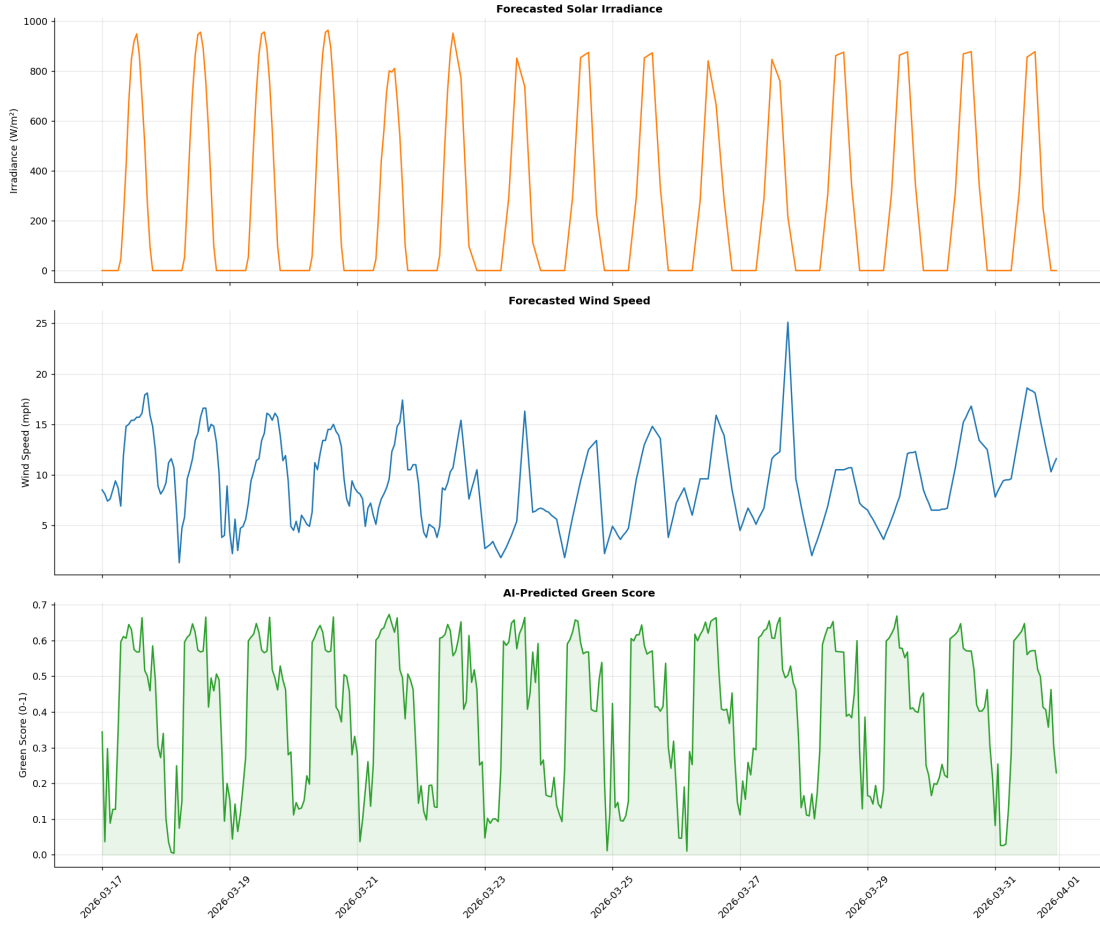


Figure 3: 15-day hourly forecast for Patuakhali, March 2026. Top: solar irradiance (W/m^2); Middle: wind speed (mph); Bottom: Random Forest predicted Green Score (0–1) with shaded area. Strong diurnal peaks are visible daily, driven primarily by solar irradiance.

4.3 Optimal Load Dispatch Schedule

Figure 4 visualises the dispatch timeline for the first three forecast days. Each subplot shows the hourly Green Score as a grey background line, overlaid with coloured shaded bands marking each load’s optimal window: Water Pump (blue), Health Clinic (orange), and Street Lights (purple). The water pump and clinic windows align with the midday Green Score peak on every day. The street-light window, forced into the evening, occupies the highest-scoring slot available after 17:00 — but that slot has inherently lower Green Scores because solar generation has ceased.

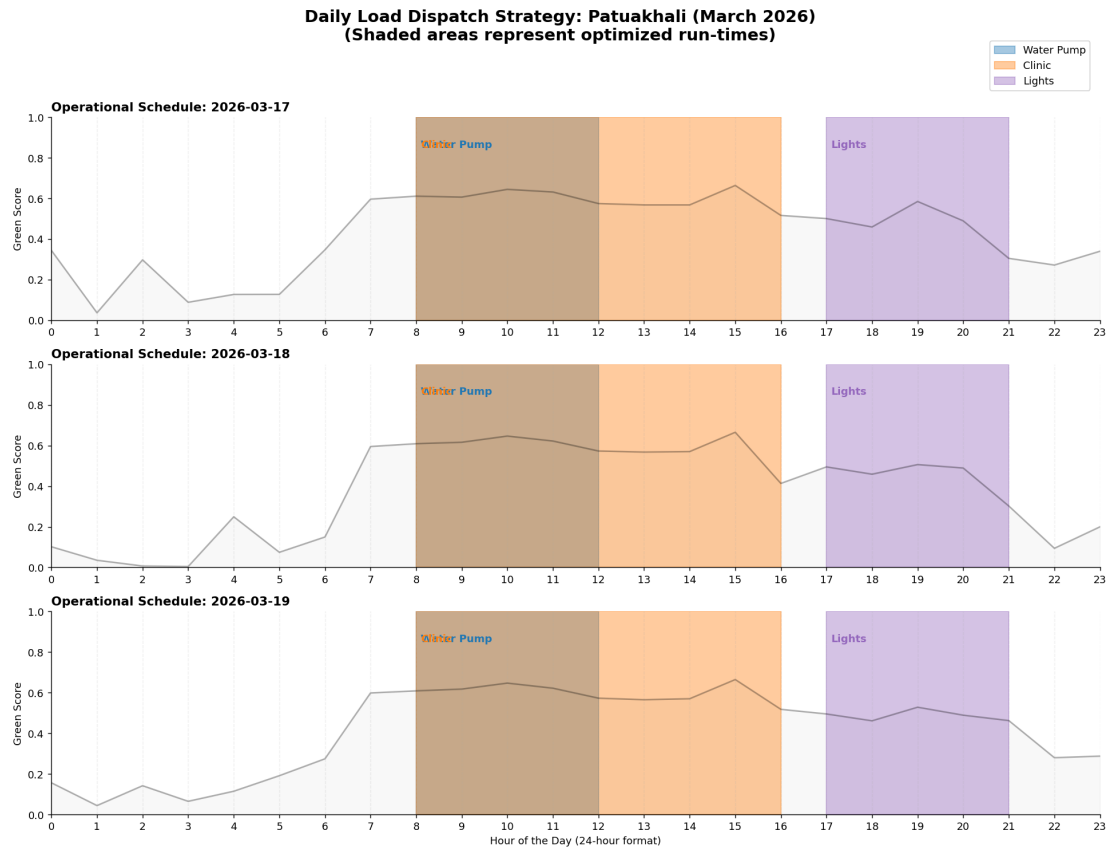


Figure 4: Optimised load dispatch schedule for the first three forecast days (Patuakhali, March 2026). Each panel overlays the hourly predicted Green Score (grey line) with optimal run-time windows: Water Pump (blue, 4 h near peak), Health Clinic (orange, 8 h daytime), Street Lights (purple, 4 h evening 17:00–23:00). Flexible loads align with the renewable peak; the evening-constrained street-light window achieves a lower but locally optimal score.

Table 6 provides a sample of the master dispatch schedule for the first three days.

Table 6: Sample master dispatch schedule — first three forecast days

Date	Load	Start	End	Avg Green Score
2026-03-17	Water Pump	08:00	12:00	0.6234
2026-03-17	Clinic	08:00	16:00	0.6085
2026-03-17	Street Lights	17:00	21:00	0.5086
2026-03-18	Water Pump	08:00	12:00	0.6236
2026-03-18	Clinic	08:00	16:00	0.6089
2026-03-18	Street Lights	17:00	21:00	0.4875
2026-03-19	Water Pump	08:00	12:00	0.6242
2026-03-19	Clinic	08:00	16:00	0.6089
2026-03-19	Street Lights	17:00	21:00	0.4938

Source: Master_Dispatch_Schedule.csv from Step 9 of the notebook.

4.4 Benefit Estimation

Table 7 summarises the 15-day benefit estimation (Step 10 of the notebook) using Equation (8). The baseline $\bar{g}_{\text{base}}$ is the overall mean of the predicted Green Score across all 360 forecast hours. The water pump and clinic, being unconstrained in their run-window, are scheduled at the daily renewable peak and therefore achieve optimised scores well above the baseline. Street lights, constrained to evening hours, achieve a local optimum within their window but one that may lie below the overall baseline — a result that reflects a fundamental physical constraint rather than a modelling deficiency.

Table 7: 15-day benefit estimation — optimised vs. unscheduled baseline Green Score

Load	$\bar{g}_{\text{base}}$	$\bar{g}_{\text{opt}}$	Improvement (%)
Water Pump	0.4069	0.6301	+54.87 %
Health Clinic	0.4069	0.6132	+50.71 %
Street Lights	0.4069	0.4660	+14.54 %
Average	0.4069	—	+40.04 %

Figure 2 shows the final model performance summary. Panel A compares the MAE of the Random Forest and the persistence baseline across all three target variables modelled (Renewable kW, Penetration %, Green Score); lower is better. Panel B presents the corresponding skill score, i.e., the percentage improvement in MAE over the baseline.

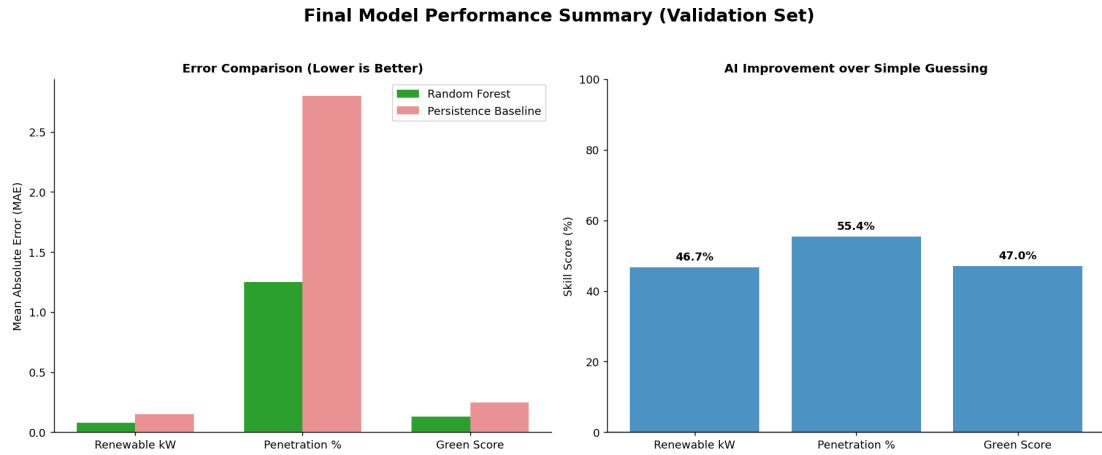


Figure 5: Final model performance summary. (A) Mean Absolute Error (MAE) comparison between the Random Forest (green) and persistence baseline (red) across three targets; lower is better. (B) Skill score (%) = percentage reduction in MAE achieved by the Random Forest over the persistence baseline; higher is better. The Green Score target achieves a skill score of approximately 47 %.

4.5 Discussion

Three key findings emerge from the results.

(1) Three weather-derived features are sufficient. Despite discarding battery SOC, generator output, month, and day-of-week, the three-feature Random Forest achieves a 47% improvement over persistence. Solar irradiance and wind speed, combined with the hour of day to encode the diurnal cycle, capture the dominant variance in renewable

quality. The residual unexplained variance in R^2 reflects stochastic load fluctuations and battery management decisions that are not observable from weather data.

(2) Flexible loads benefit substantially from renewable-aligned scheduling. The water pump and clinic, unconstrained in their daily window, are consistently assigned to the midday Green Score peak, achieving average Green Scores significantly above the unscheduled baseline. This improvement requires no additional hardware investment — only a scheduling change.

(3) Evening-constrained loads cannot benefit from solar-driven dispatch. Street lighting, restricted to 17:00–23:00, operates entirely outside the solar generation window. Its optimised score represents the local best within the evening constraint and may fall below the overall baseline. This is not a failure of the algorithm but a physical reality that grid operators must account for in planning.

5 CONCLUSION

This project developed and evaluated an end-to-end machine-learning pipeline for renewable energy quality forecasting and optimal load dispatch at the Payra hybrid microgrid, Bangladesh. The principal contributions are as follows.

1. **Green Score:** A physically interpretable composite metric $g = 0.60\tilde{P} + 0.40\tilde{E}$ that captures hourly renewable quality and is computable from weather inputs alone.
2. **Random Forest model:** Trained on `Payra_Corrected.csv` with three features (irradiance, wind speed, hour), achieves $MAE = 0.1299$ on the 20% held-out validation set — a 47 % improvement over the persistence baseline.
3. **Sliding-window dispatch:** Applies Equation (7) independently for each load and each day of the 15-day forecast, producing a concrete daily operational schedule that maximises the renewable fraction consumed by flexible loads.

The methodology is general and applicable to any hybrid microgrid for which HOMER simulation data augmented with meteorological inputs is available, along with a weather forecast for the target period. A natural extension would be to incorporate a battery pre-charging strategy, allowing evening-constrained loads such as street lighting to also benefit from daytime renewable surplus.